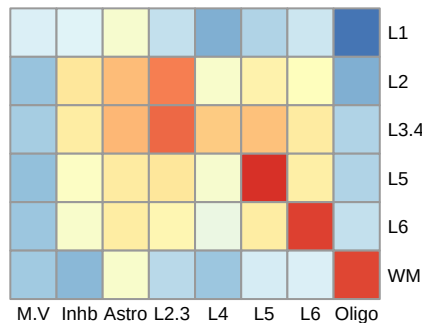
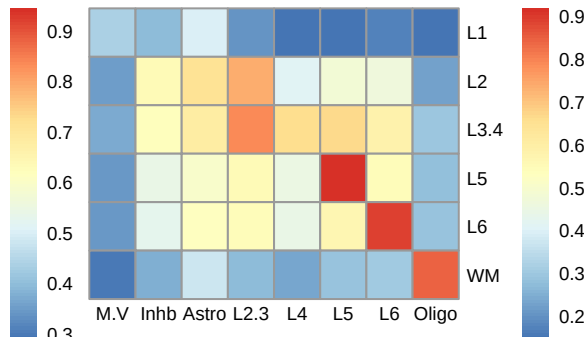


t statistic correlation between L-R annotations

NTC.MDD F



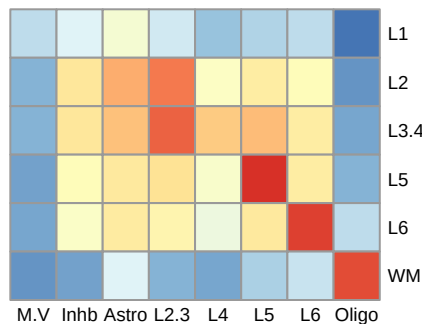
NTC.MDD M



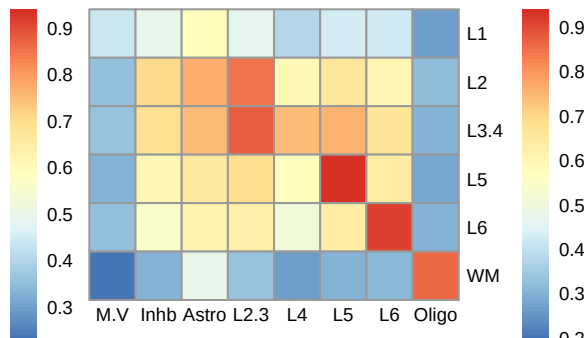
L5, L6, and WM/Oligo have extremely similar L-R results from the two annotations, consistent with the high degree of spot-level overlap between each pair of annotations.

There is a low degree of overlap between PRECAST L1 L-R results and Seurat label Micro.Vasc or Astro L-R results, despite some overlap in spot-level annotations.

NTC.BPD F

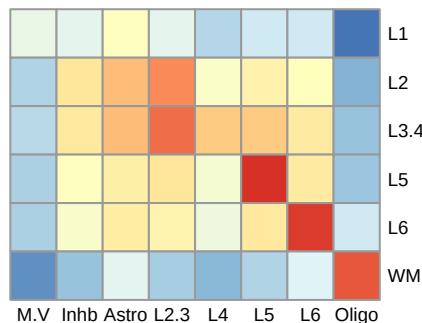


NTC.BPD M

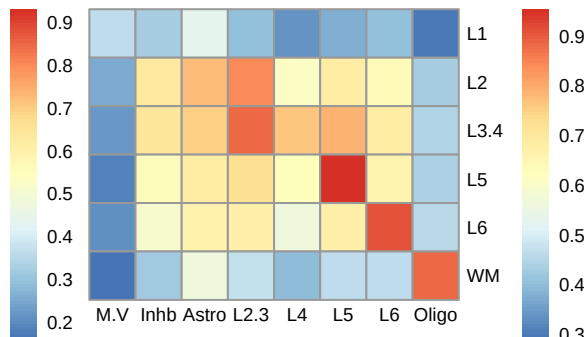


There is no clean way to look at the supra-granular comparisons between the PRECAST and Seurat label annotations. Both from the biological justification of overlapping importance of each of L2 vs L2.3 and L3.4 vs L2.3 and L3.4 vs L4, as well as from looking at the correlation of the t-statistics of the L-R results.

MDD.BPD F



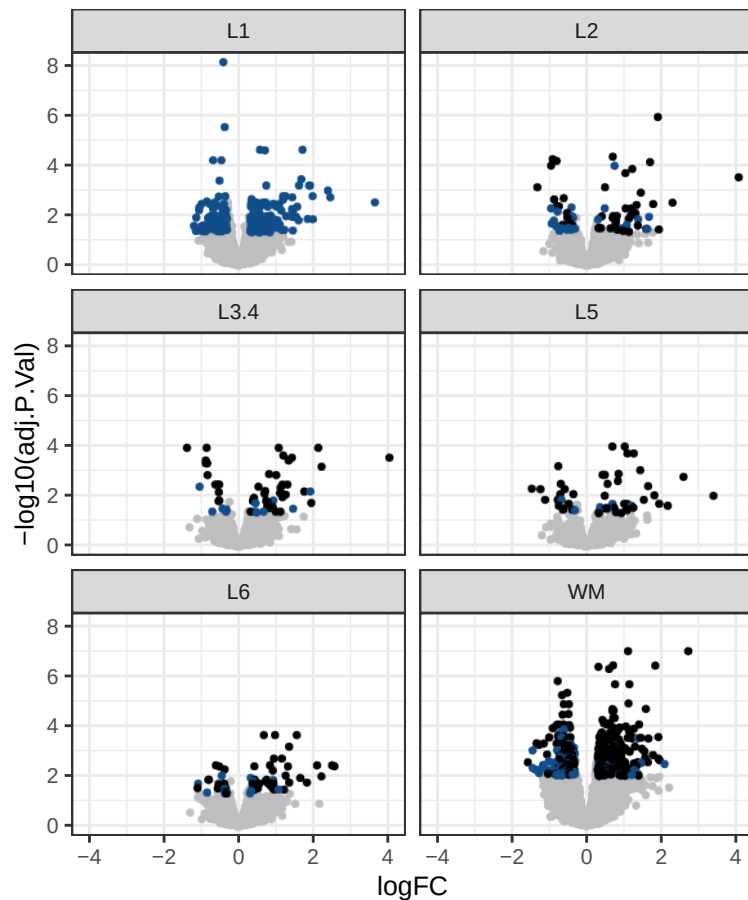
MDD.BPD M



Altogether, we will look at a combined group of L2,L3,L4 for L-R DEG consistency, as well as the 3 separate groups of L5, L6, and WM.

PRECAST smoothed L-R results

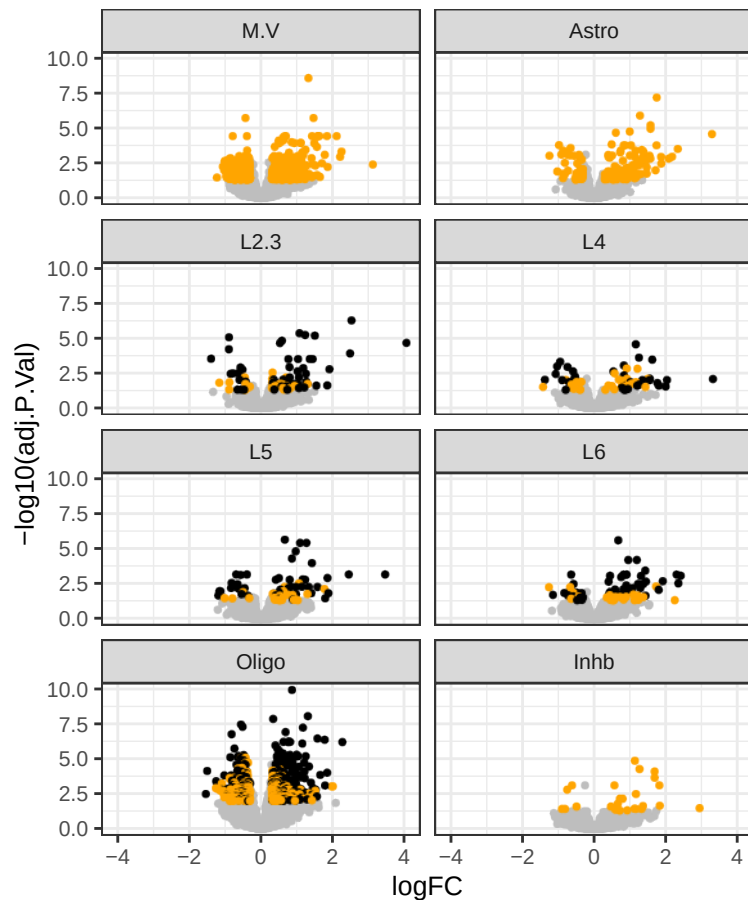
(age, PC3 covar)



• NS • PRECAST only • Paired DEG

Seurat labels L-R results

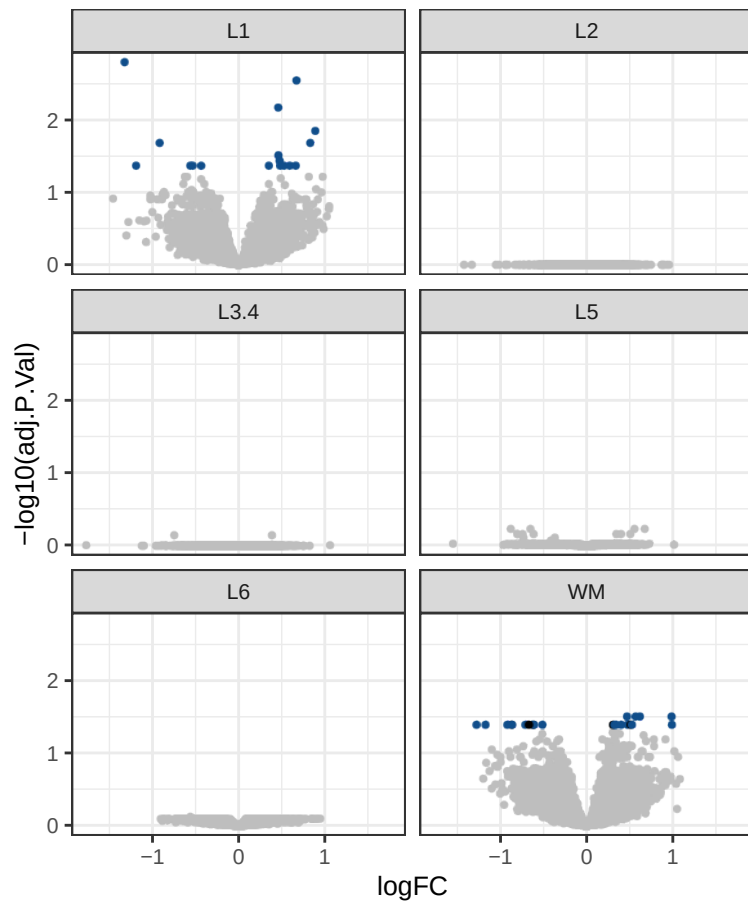
(age, PC3 covar)



• NS • Seurat only • Paired DEG

PRECAST smoothed L-R results

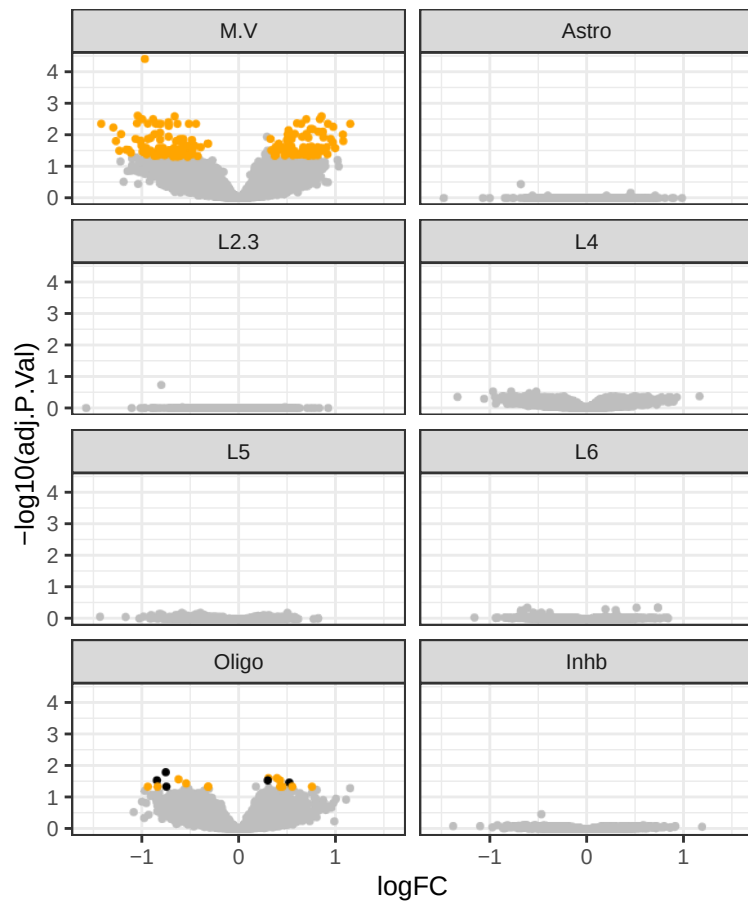
(age, PC3 covar)



● NS ● PRECAST only ● Paired DEG

Seurat labels L-R results

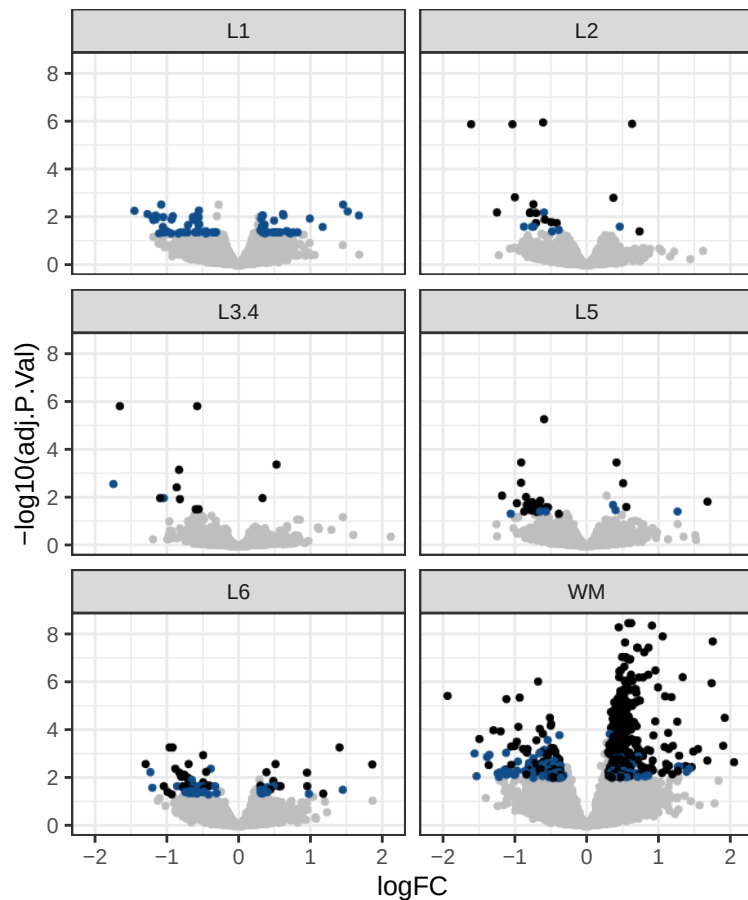
(age, PC3 covar)



● NS ● Seurat only ● Paired DEG

PRECAST smoothed L-R results

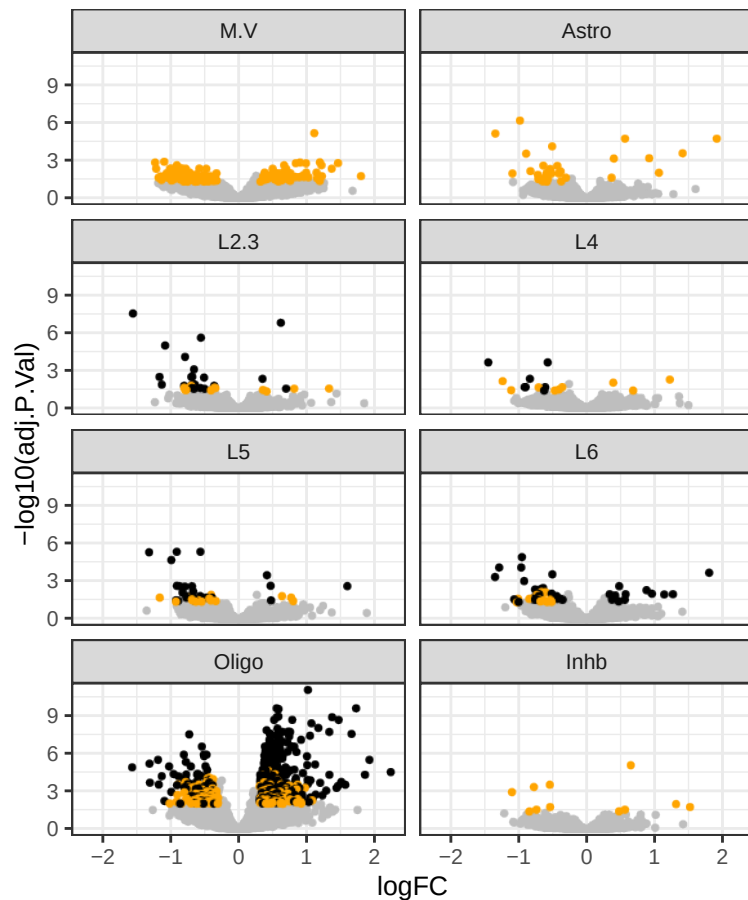
(age, PC3 covar)



● NS ● PRECAST only ● Paired DEG

Seurat labels L-R results

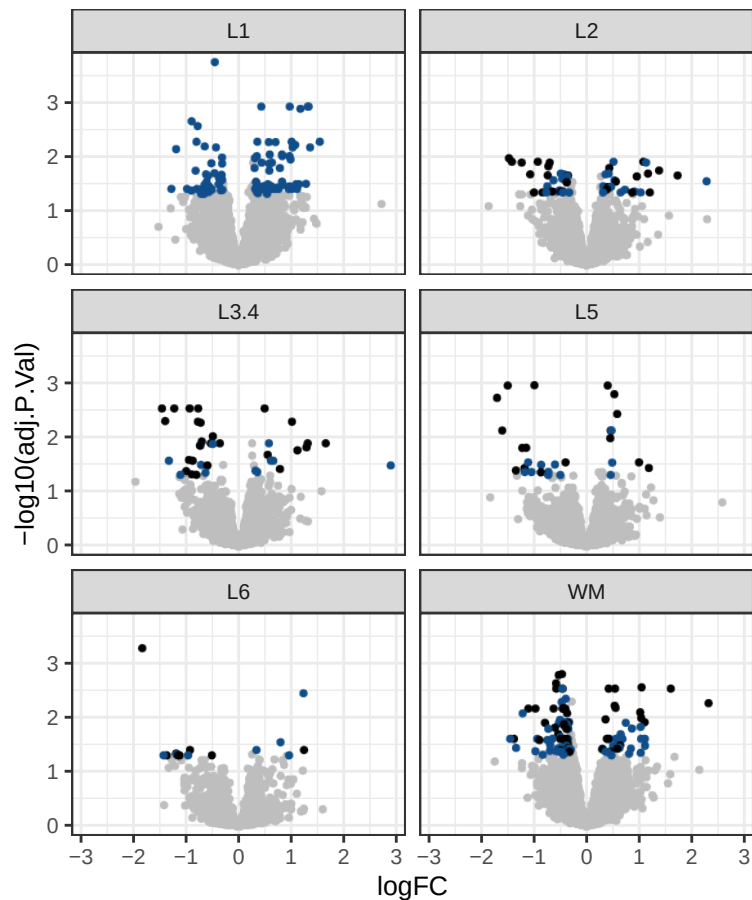
(age, PC3 covar)



● NS ● Seurat only ● Paired DEG

PRECAST smoothed L-R results

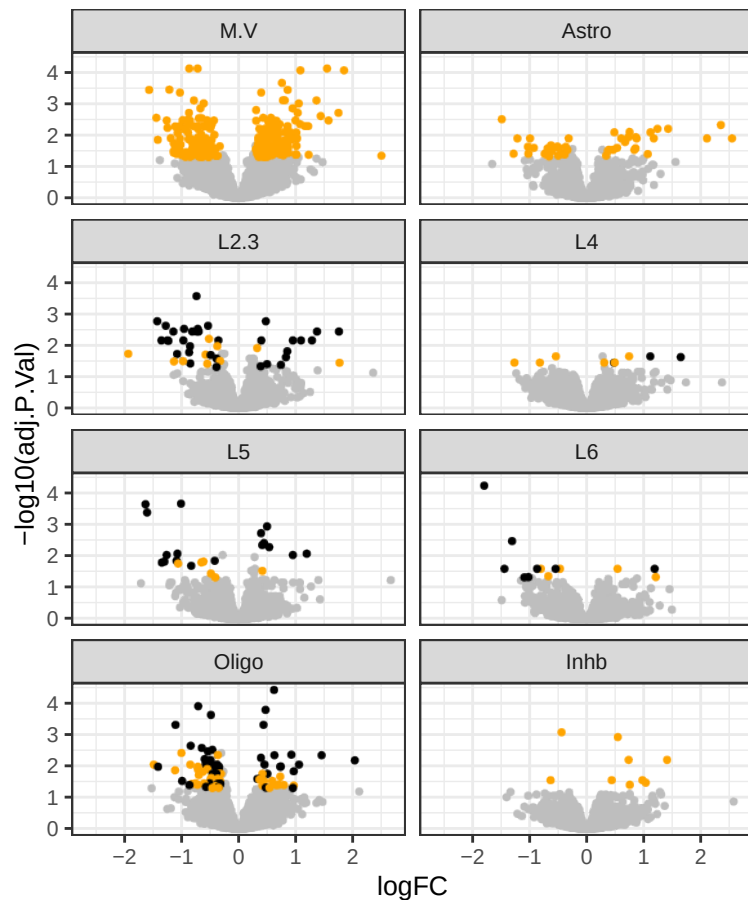
(age, PC3 covar)



● NS ● PRECAST only ● Paired DEG

Seurat labels L-R results

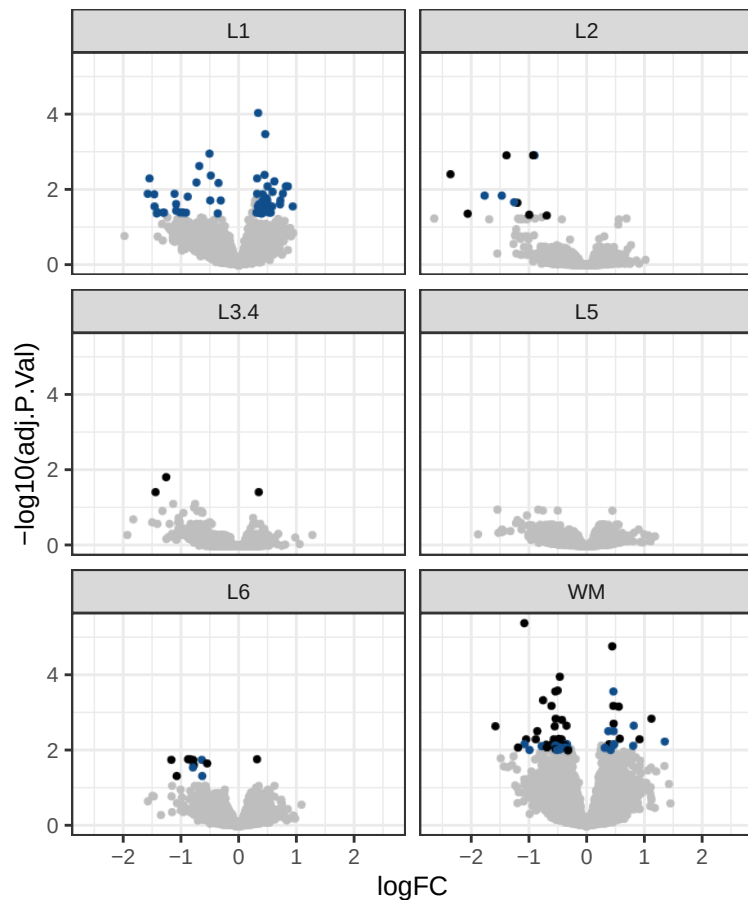
(age, PC3 covar)



● NS ● Seurat only ● Paired DEG

PRECAST smoothed L-R results

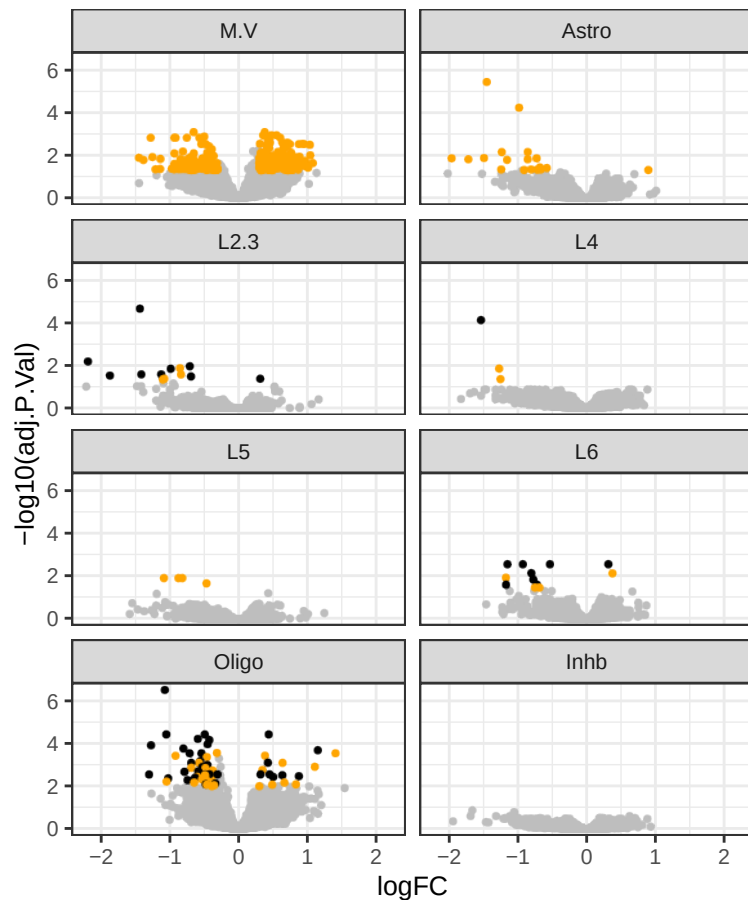
(age, PC3 covar)



● NS ● PRECAST only ● Paired DEG

Seurat labels L-R results

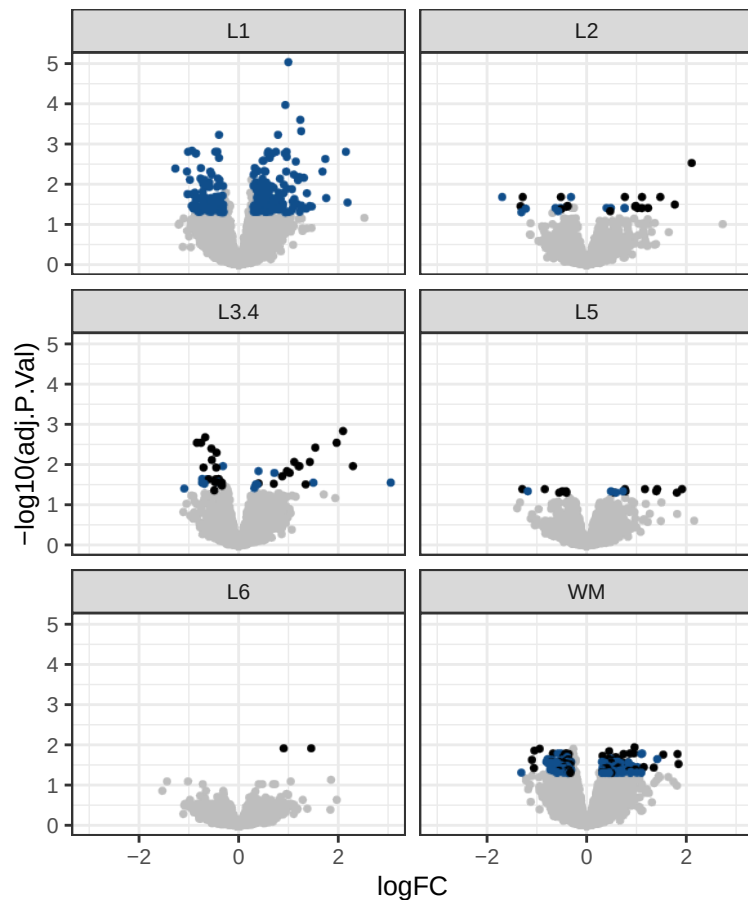
(age, PC3 covar)



● NS ● Seurat only ● Paired DEG

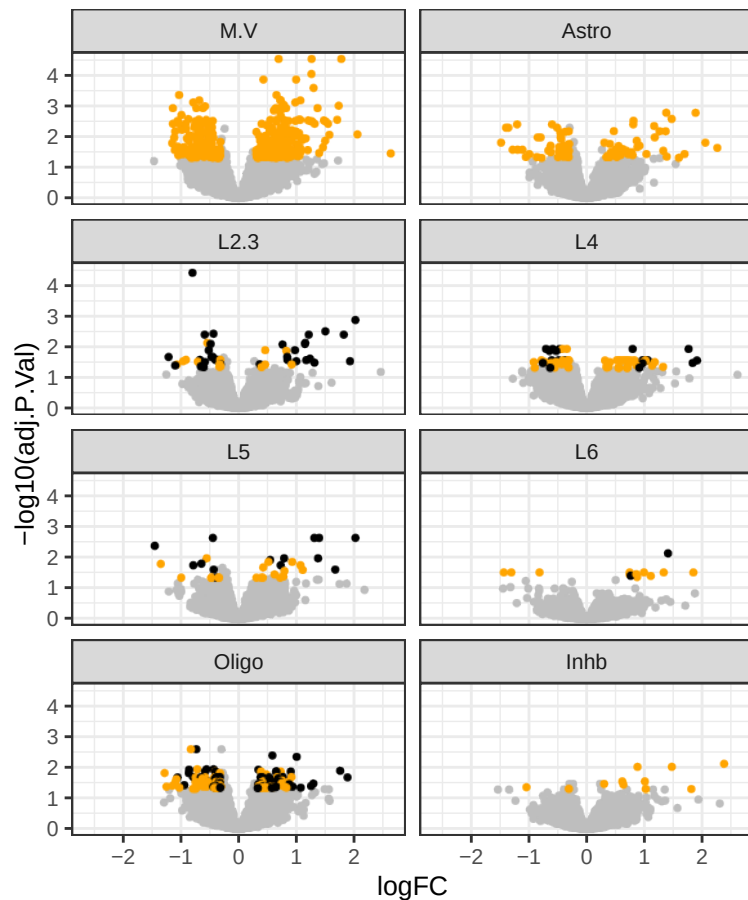
PRECAST smoothed L-R results

(age, PC3 covar)



Seurat labels L-R results

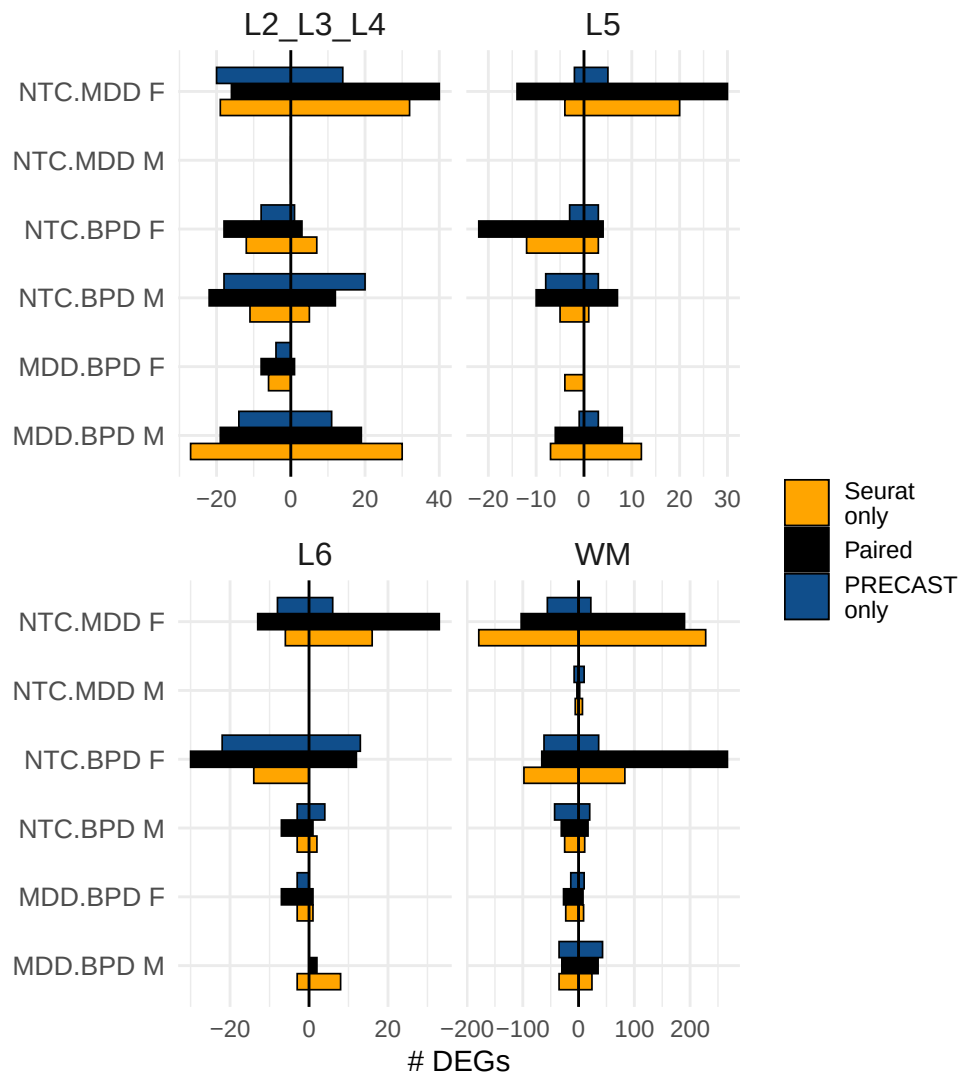
(age, PC3 covar)



● NS ● PRECAST only ● Paired DEG

● NS ● Seurat only ● Paired DEG

Number of paired L-R DEGs across matching groups

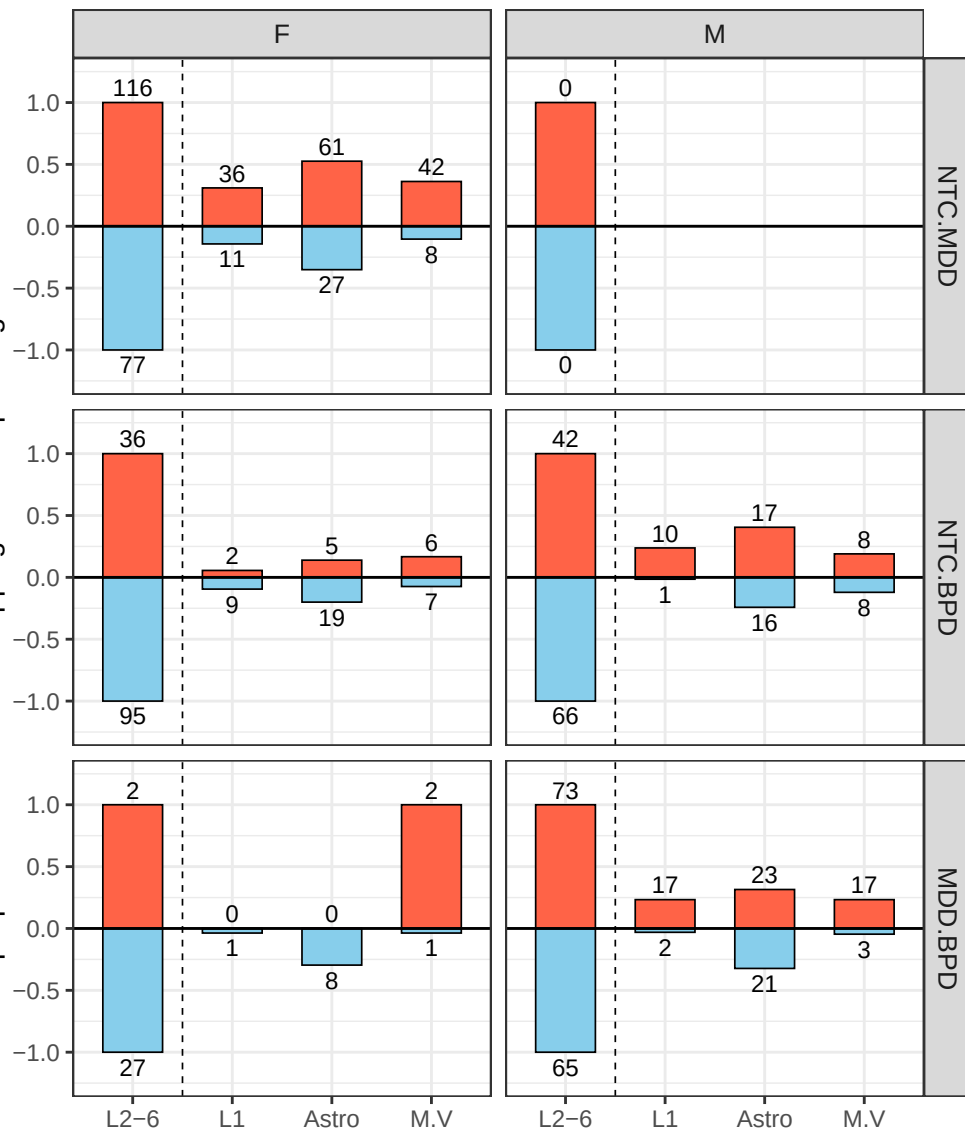


Each facet contains the data for one of the 4 groups chosen for paired L-R analysis. The y-axis contains the dx*sex comparison being examined and the x-axis indicates the number of DEGs past our significance threshold, with the positive and negative values indicating if the DEG was increased or decreased, respectively.

The fill color indicates if the DEG was significant in only one annotation model or if the DEG was significant in the same direction in both annotation models. Fill color also matches the schematic.

These plots show that there is often considerable agreement between the DEGs of the two annotation strategies. However, there are also plenty of DEGs that were only significant in one model that we likely do not want to discard entirely.

prop. of L2-6 DEGs overlapping with un-paired glial DEGs



This plot examines the overlap between any neuronal (L2-L6) DEGs from either annotation and the glial DEGs of L1 (PRECAST), Micro.Vasc (Seurat), and Astro (Seurat).

For each dx*sex comparison (facets), the y-axis indicates the proportion of L2-6 DEGs that were also DEGs in the x-axis groups. Positive/ negative values indicate whether DEGs were increased/ decreased, respectively. To help orient, the vertical dashed line separates the gene set being used in the denominator (on the left; L2-6 DEGs), versus the glial DEGs being used in the numerator (on the right).

The total number of DEGs in each bar is given by the text value, to help discern when a large proportion of L2-6 DEGs being identified as glial DEGs is due to low overall number of L2-6 DEGs. These values are also direction (increased/ decreased) specific.

NTC.MDD F

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			97	29	17	16	21	159
<i>PRECAST only</i>				15	5	2	8	56
<i>Paired DEG</i>				16		14	13	103
<i>Seurat only</i>				10	9	4	6	179
<i>Seurat DEG</i>	5	257	40	23	20	18	19	282
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			135	39	38	35	39	212
<i>PRECAST only</i>				8	7	5	6	22
<i>Paired DEG</i>				40		30	33	190
<i>Seurat only</i>				17	19	20	16	228
<i>Seurat DEG</i>	20	312	99	57	41	50	49	418
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

NTC.MDD M

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			6	0	0	0	0	11
<i>PRECAST only</i>				0	0	0	0	8
<i>Paired DEG</i>				0		0	0	3
<i>Seurat only</i>				0	0	0	0	6
<i>Seurat DEG</i>	0	92	0	0	0	0	0	9
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			11	0	0	0	0	12
<i>PRECAST only</i>				0	0	0	0	10
<i>Paired DEG</i>				0		0	0	2
<i>Seurat only</i>				0	0	0	0	7
<i>Seurat DEG</i>	0	77	0	0	0	0	0	9
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

NTC.BPD F

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			39	19	10	25	52	128
<i>PRECAST only</i>				6	2	3	22	62
<i>Paired DEG</i>				18		22	30	66
<i>Seurat only</i>				7	6	12	14	98
<i>Seurat DEG</i>	6	83	22	25	13	34	44	164
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			30	4	2	7	25	303
<i>PRECAST only</i>				1	0	3	13	36
<i>Paired DEG</i>				3		4	12	267
<i>Seurat only</i>				4	3	3	0	83
<i>Seurat DEG</i>	5	60	7	7	3	7	12	350
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

NTC.BPD M

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			36	26	24	18	10	74
<i>PRECAST only</i>				12	6	8	3	43
<i>Paired DEG</i>				22		10	7	31
<i>Seurat only</i>				8	3	5	3	25
<i>Seurat DEG</i>	2	122	21	30	3	15	10	56
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			65	25	14	10	5	37
<i>PRECAST only</i>				14	6	3	4	20
<i>Paired DEG</i>				12		7	1	17
<i>Seurat only</i>				2	3	1	2	11
<i>Seurat DEG</i>	7	142	22	14	6	8	3	28
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

MDD.BPD F

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			23	11	2	0	10	41
<i>PRECAST only</i>				4	0	0	3	14
<i>Paired DEG</i>				8		0	7	27
<i>Seurat only</i>				4	2	4	3	23
<i>Seurat DEG</i>	0	121	17	12	3	4	10	50
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			32	0	1	0	1	18
<i>PRECAST only</i>				0	0	0	0	10
<i>Paired DEG</i>				1		0	1	8
<i>Seurat only</i>				0	0	0	1	9
<i>Seurat DEG</i>	0	157	1	1	0	0	2	17
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

MDD.BPD M

Decreased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			88	13	23	7	0	65
<i>PRECAST only</i>				7	7	1	0	35
<i>Paired DEG</i>				19		6	0	30
<i>Seurat only</i>				9	18	7	3	35
<i>Seurat DEG</i>	2	139	40	25	28	13	3	65
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo

Increased

			L1	L2	L3.4	L5	L6	WM
<i>PRECAST DEG</i>			130	15	21	11	2	78
<i>PRECAST only</i>				4	7	3	0	43
<i>Paired DEG</i>				19		8	2	35
<i>Seurat only</i>				5	26	12	8	24
<i>Seurat DEG</i>	9	205	43	21	34	20	10	59
	Inhb	M.V	Astro	L2.3	L4	L5	L6	Oligo